KRISH JOSHI

Agentic Software Engineer | Autonomous AI Systems & Deterministic Test Harnesses

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PROFESSIONAL SUMMARY

Forward-deployed **Agentic Software Engineer** and 3rd-year CS undergraduate who builds production software and evaluation systems by directing autonomous AI coding agents (Google Antigravity, Claude Code, OpenAI Codex). Delivers complex multi-file engineering tasks 5–10x faster than traditional manual workflows through spec-driven development, Model Context Protocol (MCP) tool orchestration, and deterministic test harnesses (Docker sandboxing, PyTest, SHA-256 assertions) that eliminate hallucinations and prevent regressions.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Agentic Orchestration & Autonomy:** Google Antigravity, Claude Code / Opus, OpenAI Codex, Multi-Agent Loops, Subagent Delegation, Spec-Driven Development, Context Engineering
- **Verification, Testing & Evals:** Deterministic Test Harnesses, Two-Tier PyTest Frameworks, Container Sandboxing (Docker, Harbor Format), CTRF Test Schemas, SHA-256 State Assertions
- **Tool Integration & Trajectories:** Model Context Protocol (MCP), Function Calling, POSIX Shell Automation, Trajectory Auditing (Grounding, Loop Breaking, Rate Limit & 401/403/429 Handling)
- **Languages & Infrastructure:** Python (PyTest, NumPy), TypeScript/JavaScript, Bash/Linux Systems, Docker Containers, Git/GitHub CI/CD, REST APIs, JSONL Trajectory Logging

PROFESSIONAL EXPERIENCE

Handshake — Frontier AI Evaluation & Trajectory Engineering **July 2026 – Present**

AI Benchmark Developer & Agent Trajectory Auditor (Remote Contractor) *$1,267.62 Earned | 100% Quality Score*

- **Containerized Agent Evaluation (Project Dynamo, Jul–Sep 2026):** Architected and audited 10 Dockerized evaluation environments in Harbor / Terminal-Bench format to benchmark frontier models on repository-scale refactoring and GPU kernel optimization tasks.
- **Deterministic Two-Tier Verification:** Designed two-tier PyTest evaluation harnesses and cryptographic SHA-256 state assertions to prevent agent false passes, catching subtle model regressions and command execution bugs.
- **Multi-Turn Agent Trajectory Auditing (Project Octahedron, Oct 2026):** Logged 27+ hours analyzing 32 multi-turn agent reasoning and tool-calling trajectories across frontier LLMs, diagnosing tool-call grounding errors, context window saturation, and unhandled API 401/403/429 loops.
- **Rigorous Benchmark Calibration:** Audited task pass rates across multiple model families, tuning difficulty to reliably expose failure modes in planning, file editing, and test execution.

Autonomous AI Systems & Eval Lab (Open Source) **2026 – Present**

Lead Systems Developer & AI Workflow Architect *github.com/KR-007J/ai-systems-eval-portfolio*

- **Autonomous Feature Delivery Workflows:** Architected spec-driven development pipelines where autonomous coding agents (Claude, Antigravity, Codex) plan, execute, and verify tasks against executable test specifications, reducing delivery turnaround by 75%.
- **Sandbox Isolation & Security Controls:** Built safe execution environments utilizing non-root Docker sandboxes and strict path sanitization (assert_safe_path) to allow agents to safely run shell commands and modify codebases with zero host leakage.
- **Open-Source Evaluation Portfolio:** Authored and published a public portfolio showcasing reproducible Docker test environments, passing unit tests, CTRF schema reporting, and trajectory auditing rubrics under MIT license.

FEATURED ENGINEERING PROJECTS

Automated Dockerized Agent Evaluation Harness **Python, Docker, PyTest, CTRF Schema, Harbor Standards**

- Built an automated testing engine that provisions ephemeral Docker containers, injects agent-generated multi-file patches, and executes two-tier verifications against SHA-256 state assertions.
- Integrated CTRF JSON schema test reporting for automated regression detection; eliminated stochastic pass flukes.

Multi-Turn Agent Trajectory Debugger & Recovery Probe **Python, JSONL, Model Context Protocol (MCP), REST APIs**

- Developed trajectory parsing scripts to inspect multi-turn agent interaction logs, measuring tool selection accuracy and argument drift.
- Simulated synthetic API rate limits and network degradation to benchmark and harden agent recovery loops across multi-step tasks.

EDUCATION & PROFESSIONAL CERTIFICATIONS

Bachelor of Technology (B.Tech) in Computer Science & Engineering **July 2024 – June 2028 (Expected)**

Institute of Advanced Research (IAR) — Gandhinagar, Gujarat, India **Current Status: 3rd-Year Undergraduate**

- **Google Cybersecurity Professional Certificate:** Linux security controls, asset defense, vulnerability triage, Python automation.
- **NVIDIA Introduction to Networking:** Modern network architectures, low-latency communication, distributed systems interconnects.